

TrES-3b

R-0150

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230318 · created 2026-07-16T09:42:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460021.9486 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.05
Transit depth (Rp/Rs)^2	1.21 +/- 1.09 [%]
Orbital Inclination (inc)	82.24 +/- 2.27
Airmass coefficient 1 (a1)	1.0656 +/- 0.0071
Airmass coefficient 2 (a2)	-0.062 +/- 0.044
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	#2 - [61, 72]
Optimal Aperture	4.11
Optimal Annulus	17.56
Transit Duration (day)	0.025 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.11 $\pm$ 0.05 vs 0.16309 (deviation 1.04 σ)
Duration fit vs published	0.025 d vs 0.0567 d (deviation 2.39 σ)
Mid-time O-C	-4.9 min
Depth SNR	2.2
Residual scatter	0.66 %

Light curve

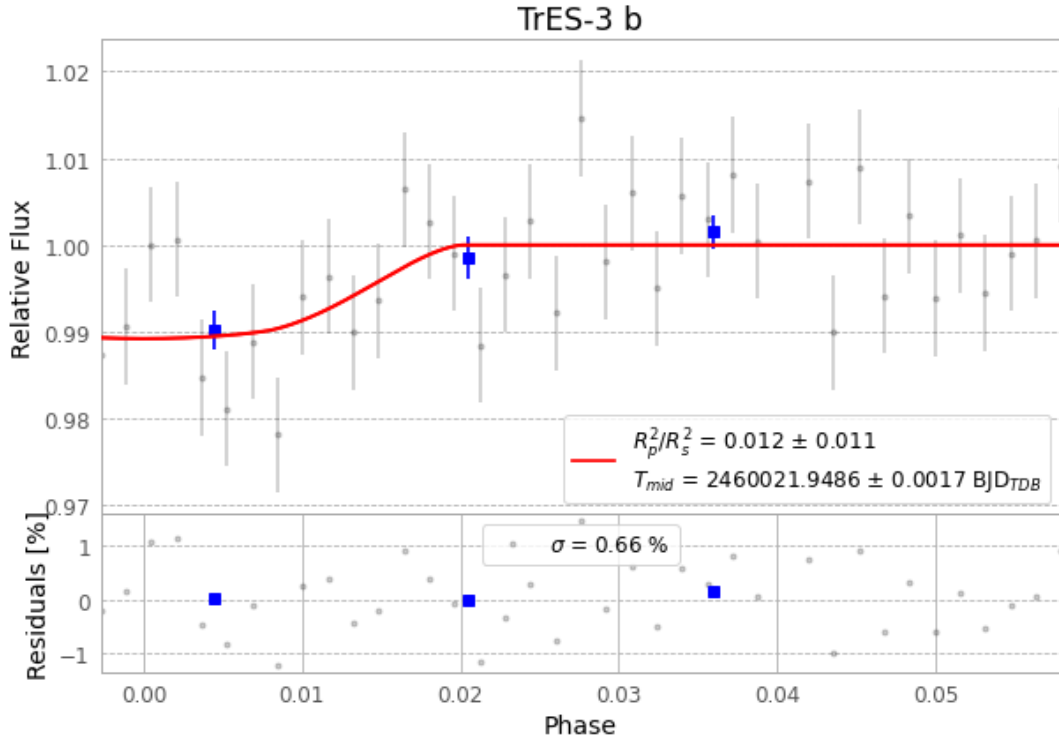


Figure 1 — FinalLightCurve_TrES-3 b_2023-03-18.png (SHA-256 17e1f7934d2a9355...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 142], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a71c2d84feac724d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

39 FITS frames (26 MB), TRES-3230318103915.FITS → TRES-3230318123315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230318.log (9c03b5986248...), FinalLightCurve_TrES-3 b_2023-03-18.png (17e1f7934d2a...), FinalLightCurve_TrES-3 b_2023-03-18.csv (b88c10837d38...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 09:42Z.